

Piper Inns Hall

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Wellington, New Zealand

PROJECTS

- **Ray Tracing – a CPU-Based Path Tracer** November 2025 – December 2025
Tools: Rust [@piperinnshall/raytracing](#)
 - Built a path tracer in Rust with ray-sphere intersection and multi-sample anti-aliasing.
 - Implemented physically-based Lambertian diffuse, Dielectric, and Metal material types.
- **Multi-Order Type Order Book** May 2025 – November 2025
Tools: Java [@piperinnshall/orderbook](#)
 - Built a limit order book matching engine supporting multiple order types with price-time priority execution.
 - Implemented order insertion and cancellation with indexed lookup and price-level data structures.
 - Designed trade matching logic handling partial fills and deterministic execution across bid / ask markets.
- **processing.nvim – a Neovim plugin for Processing integration** September 2024 – Present
Tools: Lua, Neovim, LuaRocks, GitHub Actions [@sophieforrest/processing.nvim](#)
 - Co-maintainer of a Neovim plugin providing Processing support with LSP integration.
 - Published on LuaRocks with over 100 downloads and automated version releases.
 - Implemented CI/CD pipelines using GitHub Actions, including documentation and testing workflows.

EXPERIENCE

- **Victoria University of Wellington** June 2025 – October 2025
Tutor – CGRA151 Wellington, New Zealand
 - Tutored an Introduction to Computer Graphics and Games class of 200+ students.
 - Hosted tutorials and assisted with in-person marking sessions for assignments and labs.
- **Project Kororā – Student-Run Satellite Program** April 2025 – Present
Lead Software Engineer, Git Repository Maintainer Wellington, New Zealand
 - Developed onboard software for a student-led CubeSat mission using the PyCubed flight software stack.
 - Designed CI/CD and automated testing, including Anaconda environments and linting for the CubeSat repo.
- **Privately Employed** December 2023 – Present
Elder Caregiver Wellington, New Zealand
 - Provide in-home care and mobility support for a wheelchair user alongside full-time study.

EDUCATION

- **Victoria University of Wellington** Feb 2024 – October 2026
Bachelor of Science – Computer Science & Computer Graphics (Cyber Security specialization) Wellington, New Zealand

AWARDS & CERTIFICATIONS

- **Harvard's CS50x Introduction to Computer Science** December 2022
Harvard University 100aa7725c234c51b6c3dd29090b6f3e
- **Leadership Summit Scholarship Recipient** July 2023
Festival for the Future

VOLUNTEER EXPERIENCE

- **Victoria University of Wellington Students' Association** Class Representative 2025
- **Relay For Life:** Volunteer & Fundraiser 2020 & 2023

SKILLS

- **Programming Languages:** Java, Rust, Lua, Python, GDScript, JavaScript, C
- **Systems & Performance:** Linux/Unix, Git, Networking, Algorithm Design, Data Structures, Debugging (LLDB)
- **Computer Graphics & Imaging:** Godot, Ray Tracing, Shader Programming, Lighting Models
- **Software Engineering & Tools:** CI/CD, GitHub, GitLab, Testing, AWS, SQL

INTERESTS

- **Basketball:** Selected for Wellington basketball regional team, NZ Basketball Academy Vegas tour. US Ambassadors Hoop Club Invitational Runner Up, A Grade Regional Championship Runner Up
- **Algorithmic Problem Solving:** Studying Theory of Computation and Algorithms beyond university curriculum.
- **Physical Fitness & Well-being:** Reading (Technical & Non-Technical), Tramping, Running, Gym, Walking